Epidemiology of dental diseases

Measures of frequency, oral health indices, WHO survey methods, global burden and risk factors

Practical Class 9 · Dental Public Health · 2026/2027

Chief Assistant Professor Kostadin Kostadinov, MD, PhD

Department of Social Medicine and Public Health · Medical University of Plovdiv

What you will be able to do after this class

1. Link the measures of frequency from Class 8 to dental indices.
2. Calculate and interpret **DMFT/DMFS**, **dmft/dmfs**, **SiC**, the Care Index, **CPI**, **PI/GI**, **OHI-S** and Dean's **Community Fluorosis Index**.
3. Explain why WHO uses **index ages** and a pathfinder sample.
4. Describe the global burden of oral diseases and Bulgaria's position.
5. Link sugars, fluoride, tobacco and HPV to specific diseases and measures.

Plan of the class (90 min)

0. Recap — 4 min
1. Oral health indices — 48 min
2. WHO survey methods — 8 min
3. Burden: the world and Bulgaria — 8 min
4. Risk factors and the social gradient — 9 min
5. Problems to solve — 11 min

Recap of Class 8: what the indices measure

Measure	In dental epidemiology
prevalence	share with untreated caries (D $>$ 0); share of adults with a pocket $\geq$ 6 mm. DMFT $>$ 0 = **caries-experience** prevalence
cumulative incidence	share of caries-free children who develop caries within 3 years
incidence rate	new carious surfaces per 100 person-years
mean index value	mean DMFT, PI — severity (intensity); for CPI (ordered codes) — distribution of codes

Caries experience is cumulative

Once affected, a tooth stays D, M or F. Therefore **mean DMFT** describes the accumulated caries experience on the day of examination (a cross-sectional measure), whereas the **DMFT increment** shows newly accumulated caries experience. It misses some new lesions: recurrent caries on a filled tooth (F $\longrightarrow$ D) leaves DMFT unchanged.

Every index needs three specifications: age, diagnostic threshold and examination protocol.

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1. **Oral health indices**
2. WHO methods for oral health surveys
3. Global burden and Bulgaria
4. Risk factors and the social gradient
5. Problems

What makes a good index and why we calibrate examiners

An index is a **standardised numerical scale** for a condition in a population. A good index is:

- **valid** — it measures what it claims to measure;
- **reliable** — reproducible between and within examiners;
- **simple and quick** — usable in field conditions;
- **quantitative** — allows means and comparison;
- **sensitive** to change and **acceptable** to those examined.

Calibration

Before a survey, examiners assess the same subjects and their agreement is expressed as **Cohen's kappa**. Usual threshold: $\kappa \geq 0.60$; for ordered categories — weighted kappa.

Index ≠ diagnosis

Indices describe individuals and populations but do **not replace** the clinical diagnosis and treatment plan.

Burt & Eklund, 7e (2021), ch. 12; WHO Oral Health Surveys: Basic Methods, 5th ed. (2013)

DMFT / DMFS and dmft: caries experience

DMFT = number of permanent teeth that are **D** (decayed), **M** (missing due to caries), **F** (filled). It measures **caries experience** — past and present.

- **DMFT**: 0–32 (WHO counts third molars); **DMFS**: surfaces — 4 per anterior, 5 per posterior tooth, up to 148.
- **dmft** (primary teeth): 0–20; naturally exfoliated teeth are not scored as m.
- A tooth is counted **once**: filling + secondary caries = D.
- **Not counted**: sealants, crowns and bridges for non-carious reasons, implants.

WHO rule for M

Under 30 years: only teeth **extracted because of caries**.
From 30 years: all teeth lost for any reason (codes 4 and 5) — so in adults M is not a pure measure of caries-related tooth loss.

Limitations

No denominator (teeth at risk) → always by age. D, M and F have **equal weight**. M and F depend on the dentist's decisions. DMFT = 0 does not mean "caries-free" if initial lesions are not recorded.

Klein, Palmer & Knutson, 1938; WHO (2013), pp. 47–48; Burt & Eklund, ch. 14

Read the index: which dentition and what do we count?

Index	Dentition	Unit	Sum
DMFT	permanent teeth	tooth (T = teeth)	D + M + F
DMFS	permanent teeth	surface (S = surfaces)	DS + MS + FS
dmft	primary teeth	tooth	d + m + f
dmfs	primary teeth	surface	ds + ms + fs

For one patient

2 decayed + 1 extracted because of caries + 3 filled without caries = **DMFT 6**. Six teeth have caries experience; two currently have untreated caries.

For a group

Three patients have DMFT 0, 2, 4.
Mean DMFT = (0 + 2 + 4) / 3 = **2 teeth per person**. Divide by people, not by teeth.

WHO. Oral Health Surveys: Basic Methods, 5th ed. (2013), pp. 47, 73–74; teaching examples

For DMFT each tooth receives only one category

Finding on a present permanent tooth	Category	Reason
One or several carious lesions	**D**	Count the tooth once
Filling and caries, including caries on a different surface	**D**	D takes priority over F
Permanent restoration placed because of caries; no residual or recurrent caries	**F**	Treated caries
Defective filling but no caries	**F**	A defect alone does not make it D
Temporary filling — WHO (2013)	**D**	An explicit rule of this protocol
Sealant only; restoration solely for trauma or cosmetic reasons	**0 to DMFT**	No caries experience to count

Three decayed surfaces on one tooth = D = 1, not D = 3.

WHO (2013), pp. 43–47. The temporary-filling rule depends on the protocol.

An absent tooth is not automatically M

Outside the relevant DMF sum

- Unerupted permanent tooth: code 8, outside DMFT.
- Congenital absence: outside **classic** M; WHO uses code 5.
- Supernumerary tooth: outside the standard set.
- Retained primary tooth: outside the **permanent** DMFT.
- Sound tooth or sealant only: contributes 0; status is recorded separately.

Classic rule for M

Extraction **because of caries** $\longrightarrow$ M.
Extraction after trauma or for orthodontics $\longrightarrow$ outside caries-specific M.

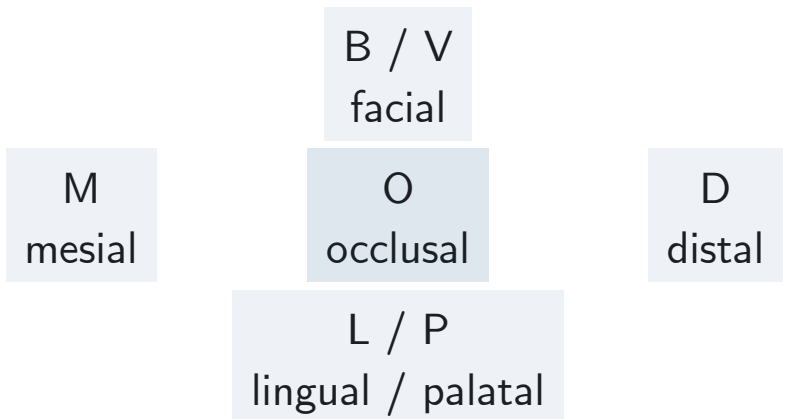
WHO (2013) age rule

Under 30: M = code 4 (caries).
From 30: M = codes 4 + 5. Code 5 includes non-caries extractions **and congenital absence**. This M therefore includes more than caries-related tooth loss.

WHO. Oral Health Surveys: Basic Methods, 5th ed. (2013), pp. 43–47

DMFS counts 4 surfaces per anterior and 5 per posterior tooth

Schematic map of a posterior tooth



Anterior = incisors and canines: no O.
Posterior = premolars and molars: with O.
M here = mesial; D = distal.

Tooth set	Maximum surfaces
Permanent, 28 teeth	$12 \times 4 + 16 \times 5 = \mathbf{128}$
Permanent, 32 teeth	$12 \times 4 + 20 \times 5 = \mathbf{148}$
Primary, 20 teeth	$12 \times 4 + 8 \times 5 = \mathbf{88}$

An extracted tooth included in M

When all its surfaces are counted: anterior $\longrightarrow$ **MS +4**; posterior $\longrightarrow$ **MS +5**. For DMFT either tooth gives M +1.

Teaching schematic and arithmetic; standard 4/5-surface DMFS convention; WHO (2013), Annexes 3–4

One patient: DMFT = 4, but DMFS = 12

Hypothetical patient aged 25; all other teeth sound. Count all surfaces of an extracted tooth.

Tooth	Finding	For DMFT	For DMFS
16	Caries on occlusal and mesial surfaces	D +1	DS +2
26	Mesial–occlusal–distal restoration; no caries	F +1	FS +3
36	Extracted because of caries	M +1	MS +5
11	Mesial caries; distal filling without caries	D +1	DS +1; FS +1

Count teeth

D = 2; M = 1; F = 1

DMFT = 2 + 1 + 1 = 4

Count surfaces

DS = 3; MS = 5; FS = 4

DMFS = 3 + 5 + 4 = 12

DMFS cannot be obtained by multiplying DMFT by 4 or 5. You need findings for each surface.

Teaching example; FDI tooth notation; DMF rules from WHO (2013)

Mixed dentition requires two separate indices

Primary teeth — example

Tooth	Finding	Surfaces
55	caries on 2 surfaces	ds +2
75	extracted due to caries	ms +5
64	filling on 2 surfaces	fs +2
51	filling on 1 surface	fs +1

All other primary teeth are sound.

dmft = 1 + 1 + 2 = 4

dmfs = 2 + 5 + 3 = 10

Permanent teeth in the same child

16 decayed; 26 filled without caries; all other erupted permanent teeth sound.

DMFT = 1 + 0 + 1 = 2.

Report: **dmft 4; DMFT 2.**

Natural exfoliation is not m

A naturally shed primary tooth is not counted as lost because of caries. If the cause is uncertain, use age and the survey protocol; do not automatically assign m.

WHO (2013), pp. 45–47; teaching example; maximum 20 primary teeth / 88 surfaces

The same DMFT can describe very different clinical situations

Patient	D	M	F	DMFT	Practical meaning
A	6	0	0	**6**	6 teeth with untreated caries
B	0	0	6	**6**	6 restored teeth without caries
C	0	6	0	**6**	6 teeth lost because of caries

What the sum does not show

Severity and activity of each lesion, an exact treatment plan or the number of teeth at risk. Initial lesions may fall below the diagnostic threshold. Sealants are recorded separately.

How to interpret it

Report **D, M and F separately**, alongside age and protocol. Calibrate examiners. With D $\longrightarrow$ F the total DMFT stays the same despite treatment.

DMF describes accumulated caries experience. Its components explain what lies behind the sum.

Teaching examples; WHO (2013); Burt & Eklund, 7e (2021), ch. 12, 14

Worked example: DMFT, SiC and Care Index in one class

Data

12 children aged 12. Individual DMFT:
0, 0, 0, 0, 1, 1, 2, 2, 3, 4, 5, 7
Group totals: D = 10, M = 1, F = 14.

SiC (Significant Caries Index)

Mean DMFT of the **third** with the highest caries experience (Bratthall, 2000). Bratthall's proposed goal: SiC < 3 in 12-year-olds by 2015.

Measure	Calculation	Result
Mean DMFT	25 / 12	**2.08**
Caries-experience prevalence (DMFT > 0)	8 / 12	**66.7%**
DMFT = 0 (at D3 threshold)	4 / 12	33.3%
SiC (top 4 children)	(7 + 5 + 4 + 3) / 4	**4.75**
Care Index 100 × ΣF / ΣDMFT	14 / 25	**56%**
Untreated share ΣD / ΣDMFT	10 / 25	40%

The mean of 2.08 looks "low" (below the goal of 3), but one third of the class has on average 4.75 affected teeth — caries is polarised.

ICDAS: caries as a process, not a “hole”

Code	Criterion (visual, after drying)
0	sound surface
1	first visual change in enamel — usually seen after air drying (in fissures sometimes also when wet)
2	distinct visual change in enamel — visible when wet
3	localised enamel breakdown without visible dentine
4	underlying dark shadow from dentine, with or without enamel breakdown
5	distinct cavity with visible dentine
6	extensive distinct cavity with visible dentine (more than half of the surface)

D1 and D3 thresholds

D1 — all stages, including initial enamel lesions.

D3 — lesions into dentine (cavitation). D1/D3 are **not** ICDAS codes; every study must state which codes it includes.

Practical consequence

At the D1 threshold, prevalence and DMFT are **higher**. Comparisons between countries and years are valid only with the **same threshold** and protocol.

Ismail AI et al. Community Dent Oral Epidemiol. 2007;35:170–178. doi:10.1111/j.1600-0528.2007.00347.x

Periodontal indices: from CPITN to the modified CPI (WHO, 2013)

CPITN — Ainamo et al. (1982)

Code	Finding	Need (TN)
0	healthy periodontium	0 — none
1	bleeding on probing	I — oral hygiene
2	calculus / retention	II — I + scaling
3	pocket 4–5 mm	II
4	pocket ≥ 6 mm	III — complex treatment

6 sextants; the sextant score is the highest; WHO probe with a 0.5 mm ball tip.

CPI modified — WHO (2013)

- **Bleeding**: all teeth present — 0 absent / 1 present.
- **Pockets** (from age 15): 0 — none; 1 — 4–5 mm; 2 — ≥ 6 mm.
- **Loss of attachment** (from age 15, by sextant, index teeth): 0 — 0–3 mm; 1 — 4–5; 2 — 6–8; 3 — 9–11; 4 — ≥ 12 mm.

Severe periodontal disease in WHO/GBD estimates: pocket ≥ 6 mm, CPI(TN) code 4 or loss of attachment > 6 mm.

Ainamo J et al. Int Dent J. 1982;32:281–291; WHO Oral Health Surveys: Basic Methods, 5th ed. (2013), pp. 47–51

Hygiene and gingival indices: PI, GI, OHI-S

Plaque Index (Silness & Löe, 1964)

0 — no plaque
1 — thin film, detected only with the probe
2 — moderate, visible to the naked eye
3 — abundant soft matter

Mean of 4 surfaces on 6 index teeth.

Gingival Index (Löe & Silness, 1963)

0 — normal gingiva
1 — mild inflammation, no bleeding
2 — moderate: redness, oedema, **bleeding on probing**
3 — severe: marked oedema, ulceration, spontaneous bleeding

0.1–1.0 mild; 1.1–2.0 moderate; 2.1–3.0 severe gingivitis.

OHI-S (Greene & Vermillion, 1964)

DI-S + **CI-S** (debris + calculus), each 0–3 on 6 surfaces: buccal of 16, 26; labial of 11, 31; lingual of 36, 46. DI-S = Σ scores / surfaces examined; CI-S likewise.

Range 0–6: **0–1.2** good; **1.3–3.0** fair; **3.1–6.0** poor hygiene.

Silness & Löe. Acta Odontol Scand. 1964;22:121–135. doi:10.3109/00016356408993968 · Löe & Silness. 1963;21:533–551. doi:10.3109/00016356309011240 · Greene & Vermillion. JADA. 1964;68:7–13

Fluorosis: Dean's index and the Community Fluorosis Index (CFI)

Category	Weight	Criterion
normal	0	smooth, glossy, pale creamy-white enamel
questionable	0.5	a few white flecks
very mild	1	opaque white areas $<$ 25% of the surface
mild	2	white areas 25–50%
moderate	3	all surfaces, wear, brown stain
severe	4	hypoplasia, pitting, corroded appearance

$$\mathrm{CFI} = \frac{\sum n_i \times w_i}{N}$$

Example (100 children)

40 normal, 20 questionable, 20 very mild, 12 mild, 6 moderate, 2 severe:
$(0 + 10 + 20 + 24 + 18 + 8) \,/\, 100 = \mathbf{0.80}$

A person is scored on the two most affected teeth; if they differ — on the **less affected** one; when in doubt — the lower category. CFI weights are not the WHO codes (0–5). Historical threshold (Dean): CFI 0.4–0.6 borderline; $>$ 0.6 — a public health problem.

Dean HT. JADA. 1934;21:1421–1426. doi:10.14219/jada.archive.1934.0220; Burt & Eklund, ch. 19; WHO (2013), pp. 51–52

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WHO: Oral Health Surveys — Basic Methods, 5th edition (2013)

Age	Why it was chosen
5 yrs	caries in primary teeth — changes quickly; school entry
12 yrs	**global indicator age**: last age easily reached through schools; most permanent teeth except third molars have usually erupted
15 yrs	permanent teeth exposed for 3–9 years; periodontium in adolescents
35–44 yrs	standard adult group: full effect of caries, severe periodontitis, care
65–74 yrs	life-course perspective; planning for older people

Pathfinder survey

Stratified cluster sample: subgroups with different disease levels (urban/rural, regions), **25–50 people** per age group at each site. Example: 12 sites × 25 = 300 people for one age (25 for low disease levels, about 50 for high). Site selection determines generalisability — this is not a national probability sample.

Standardisation

Single form, codes, WHO probe, training and calibration, repeat examinations.

WHO. Oral Health Surveys: Basic Methods, 5th ed. Geneva: WHO; 2013. ISBN 978-92-4-154864-9, pp. 14–19

From DMFT targets to universal health coverage

Quantitative goals

- **DMFT $\leq$ 3** in 12-year-olds by 2000 (FDI/WHO) — reached in many countries already in the 1990s.
- **SiC $<$ 3** in 12-year-olds by 2015 (Bratthall, 2000) — focuses attention on the most affected third.

WHO framework 2021–2030

- Resolution **WHA74.5** on oral health (2021).
- **Global strategy** on oral health (decision WHA75(11), 2022).
- **Global oral health action plan 2023–2030** (decision WHA76(9), 2023): oral health in NCD programmes and in the universal health coverage package.

The goal shifts from "mean DMFT below 3" to equal access to prevention and treatment for all groups.

Bratthall D. Int Dent J. 2000;50:378–384. doi:10.1111/j.1875-595X.2000.tb00572.x; WHO. Global strategy and action plan on oral health 2023–2030 (2024), ISBN 978-92-4-009053-8

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The global burden: the most widespread diseases of humankind

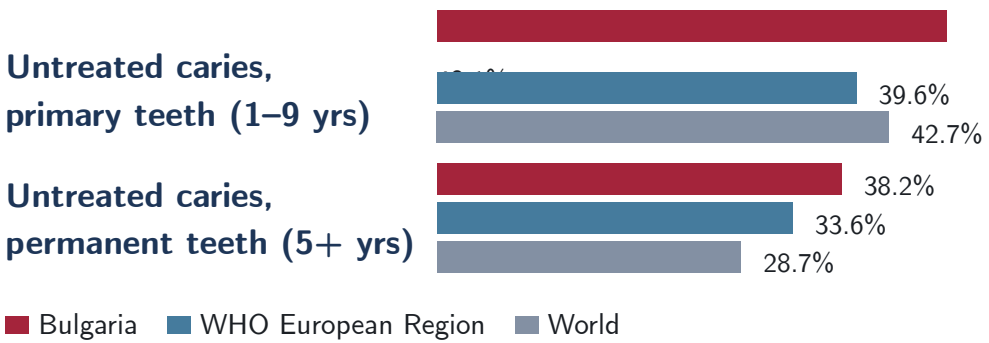
Condition (2019, GBD)	Cases	Prevalence
untreated caries, permanent teeth (5+ yrs)	≈ 2 billion	28.7%
severe periodontal disease (15+ yrs)	≈ 1 billion	18.8%
untreated caries, primary teeth (1–9 yrs)	514 million	42.7%
edentulism (20+ yrs)	350 million	6.8%
lip and oral cavity cancer (2020): **new cases / deaths**	377,713	177,757 deaths

What follows

- In total ≈ **3.5 billion** people (GBD 2019) — about 1 billion more than the cases of the five main NCDs combined.
- Cases have grown by more than 1 billion since 1990 (+50%) — mainly due to population growth and ageing.
- Spending on oral care: almost **USD 390 billion** a year.

WHO. Global oral health status report (2022), ISBN 978-92-4-006148-4, tables 2–9; GLOBOCAN 2020; periodontal and edentulism prevalence age-standardised

Bulgaria, the European Region and the world: untreated caries (GBD 2019)



Only untreated-caries indicators are compared; they are computed the same way (not age-standardised) in all three sources.

Bulgaria profile (WHO, 2022)

- Severe periodontal disease (15+): **13.9%**; edentulism (20+): **13.5%** (GBD 2019).
- 455 new cases of lip and oral cavity cancer (2020); 3.3 per 100,000 (age-standardised incidence).
- Sugar: **74.4 g/day** per person (availability, not measured intake; 2019).
- Tobacco use 15+ (2019, age-standardised): **39.4%**.
- 9,800 dentists (2019) — **14.0 per 10,000**.
- Spending on dental care: USD 16 per person (2019).

WHO. Oral Health Country Profile – Bulgaria (2022), WHO/UCN/NCD/MND/BGR/2022.1; WHO Global oral health status report (2022), tables 3, 5, 7, 9

What Bulgarian studies show

Study	Group	Main result
Künzel, 1996 (review)	12 yrs, Eastern Europe	DMFT in Bulgaria **3.1** — among the lowest in the region
Ivanova et al., 1995	6 yrs, kindergartens	fluoridated milk (1 mg F on programme days, 180–200 days/yr): dmft **40%** lower after 3.5 years; the authors note design limitations
Onov & Beltcheva, 2020	12 yrs, Plovdiv, n = 228	DMFT **1.57** in the city and **2.92** in the villages; caries-free 38.4% vs 17.5%
Kondeva et al., 2026	18 yrs, 2 municipalities, n = 480	quota sample; DMFT **7.48**: D 3.91, M 0.13, F 3.43 — D is about 52% of DMFT

Local studies show differences: urban–rural, social status and access to care are associated with who carries the burden.

Künzel W. Int J Paediatr Dent. 1996;6:221–226. doi:10.1111/j.1365-263x.1996.tb00249.x · Ivanova K et al. Adv Dent Res. 1995;9:120–121. doi:10.1177/08959374950090020701 · Onov MP, Beltcheva AB. Folia Med. 2020;62:159–164. doi:10.3897/folmed.62.e47894 · Kondeva V et al. Children. 2026;13:1087. doi:10.3390/children13081087

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Caries: sugars the main risk factor, fluoride the main protective factor

Free sugars

- Systematic review for WHO (Moynihan & Kelly, 2014): 42 of 50 studies in children found a sugar–caries association; **moderate-quality** evidence that caries is lower when intake is $<$ **10%** of energy.
- WHO guideline (2015): $<$ **10%** of energy (strong recommendation), $<$ **5%** (conditional).
- Both the **amount** and the **frequency** of intake matter.

Fluoride

- Fluoride toothpaste: prevented fraction **24%** (95% CI 21–28%) — relative reduction of the D(M)FS **increment** in children versus non-fluoride toothpaste (Cochrane, 2003; 70 trials in the meta-analysis).
- Higher concentration $\longrightarrow$ stronger effect; in young children (swallowing while teeth develop) also a higher risk of fluorosis (Walsh et al., 2019).
- Community measures: fluoridation of water, salt or milk.

Moynihan PJ, Kelly SAM. J Dent Res. 2014;93:8–18. doi:10.1177/0022034513508954 · WHO. Guideline: sugars intake (2015) · Marinho VCC et al. Cochrane 2003. doi:10.1002/14651858.CD002278 · Walsh T et al. Cochrane 2019. doi:10.1002/14651858.CD007868.pub3

Tobacco, alcohol, HPV and the common risk factor approach

Tobacco, alcohol, diabetes

Smoking is a major risk factor for **periodontitis**; diabetes (hyperglycaemia) also raises the risk. Tobacco and alcohol cause **oral cancer**, with substantially higher risk when combined. In Bulgaria 39.4% of people aged 15+ use tobacco (WHO, 2019).

HPV and oropharyngeal cancer

SEER, USA, 1988–2004: the incidence of **HPV-positive** oropharyngeal cancer rose by **225%** (from 0.8 to 2.6 per 100,000), while **HPV-negative** cancer fell by 50% (Chaturvedi et al., 2011).

Common risk factor approach

Sheiham & Watt (2000): sugar, tobacco, alcohol, hygiene and stress are shared by oral diseases, cardiovascular disease, diabetes and cancer. **Population-level** measures — taxes on sugar-sweetened beverages and tobacco, marketing regulation, healthy school environments — act on many diseases at once.

Chaturvedi AK et al. J Clin Oncol. 2011;29:4294–4301. doi:10.1200/JCO.2011.36.4596 · Sheiham A, Watt RG. Community Dent Oral Epidemiol. 2000;28:399–406. doi:10.1034/j.1600-0528.2000.028006399.x

The social gradient: evidence from the Dunedin cohort (New Zealand)

About 1000 children born 1972–1973, followed from birth; dental examinations at ages 5 and 26.

At age 26	**low child-hood SES**	**high child-hood SES**
periodontal disease (attachment loss $>$ 4 mm at $\geq$ 1 site)	**31.1%**	11.9%
$\geq$ 4 surfaces affected by caries	**32.2%**	9.9%

Prevalence at age 26 (not incidence). In this study upward social mobility **did not erase** the effect of a disadvantaged childhood (Poulton et al., 2002).

Gradient by social trajectory

From least to most disease at age 26: **high–high** → low–high → high–low → **low–low** (Thomson et al., 2004).

Oral health inequalities are stepwise, not just "poor versus rich" — measures must cover the whole gradient.

Poulton R et al. Lancet. 2002;360:1640–1645. doi:10.1016/S0140-6736(02)11602-3 · Thomson WM et al. Community Dent Oral Epidemiol. 2004;32:345–353. doi:10.1111/j.1600-0528.2004.00173.x · Watt RG et al. Lancet. 2019;394:261–272. doi:10.1016/S0140-6736(19)31133-X

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Problems to solve in class

Problem 1 · DMFT

Patient aged 35: caries on 16 and 47; fillings on 26, 36, 46; 37 and 18 were present and have been lost (cause unknown); sealant on 27; crown on 11 after trauma. Calculate DMFT.

Problem 2 · SiC

9 children aged 12: DMFT 0, 0, 1, 1, 2, 3, 3, 6, 8. Calculate mean DMFT, caries-experience prevalence and SiC.

Problem 3 · Incidence

At the start of the year a care home has 120 residents with at least one natural tooth. All are followed for 1 year; 18 become edentulous. What is the cumulative incidence of edentulism?

Problem 4 · CFI

50 children: 25 normal, 10 questionable, 10 very mild, 5 mild. Calculate the CFI and interpret it.

Answers

Problem	Solution	Answer
1	D = 2 (16, 47); F = 3 (26, 36, 46); M = 2 (37, 18 — from age 30 all missing teeth count); sealant and crown after trauma not counted	**DMFT = 7**
2	sum 24 / 9 = 2.67; caries experience 7 / 9 = 77.8%; SiC = (8 + 6 + 3) / 3	**2.67; 77.8%; 5.67**
3	18 / 120 over 1 year	**15%** over one year
4	(25×0 + 10×0.5 + 10×1 + 5×2) / 50 = 25 / 50	**CFI = 0.50** — borderline, below the 0.6 threshold

Without a denominator, a threshold and a period a number means nothing — check them in every comparison.

Summary

Measurement

1. Prevalence is a "snapshot"; incidence is a "film" of new cases among those at risk.
2. DMFT is **cumulative** caries experience; SiC shows the most affected third.
3. Compare only indices with the **same threshold**, protocol and age.
4. WHO index ages and pathfinder sampling make data comparable.

Burden and causes

1. Oral diseases affect ≈ 3.5 billion people (GBD 2019; ≈ 3.7 billion in GBD 2021).
2. Bulgaria: high untreated caries and edentulism, high tobacco use (39.4%).
3. For caries, sugar is the main risk factor and fluoride the main protective factor.
4. Inequalities are **stepwise** and begin in childhood.

Next class: **epidemiological studies**.

Thank you for your attention!